

JavaScript & Programming Aptitude Questions

1. Check if Number is Integer

```
function isInt(num) {  
    return num % 1 === 0;  
}  
  
console.log(isInt(4)); // true  
console.log(isInt(12.2)); // false  
console.log(isInt(0.3)); // false
```

2. Closure Function Example

```
function createBase(baseNumber) {  
    return function(N) {  
        return baseNumber + N;  
    }  
}  
  
var addSix = createBase(6);  
  
console.log(addSix(10)); // 16  
console.log(addSix(21)); // 27
```

3. Largest of Four Numbers (JS Version)

```
function largestFour(arr) {  
    return Math.max(...arr);  
}  
  
console.log(largestFour([10, 25, 5, 40])); // 40
```

4. argc Output Program Explanation

```
#include <stdio.h>  
  
int main(int argc, char* argv[])  
{  
    printf("%d", argc);  
    return 0;  
}
```

Input:
C:\TC\BIN>prg_1 includehelp.com C C++ Java

Output:
5

Explanation:
argc counts total command line arguments including program name.

5. Start Thread in Java

```
public class ThreadEx extends Thread  
{  
    public void run()  
    {  
        System.out.println("Running...");  
    }  
}
```

```

    }

    public static void main(String args[])
    {
        ThreadEx T1 = new ThreadEx();
        T1.start(); // start thread
    }
}

```

6. Reverse a String

```

function reverseString(str) {
    return str.split('').reverse().join('');
}

console.log(reverseString("hello")); // olleh

```

7. Check Palindrome

```

function isPalindrome(str) {
    const reversed = str.split('').reverse().join('');
    return str === reversed;
}

console.log(isPalindrome("madam")); // true

```

8. Remove Duplicate Elements

```

function removeDuplicates(arr) {
    return [...new Set(arr)];
}

console.log(removeDuplicates([1,2,2,3,4,4]));

```

9. FizzBuzz

```

for (let i = 1; i <= 20; i++) {
    if (i % 15 === 0) {
        console.log("FizzBuzz");
    } else if (i % 3 === 0) {
        console.log("Fizz");
    } else if (i % 5 === 0) {
        console.log("Buzz");
    } else {
        console.log(i);
    }
}

```

10. Factorial

```

function factorial(n) {
    if (n === 0) return 1;
    return n * factorial(n - 1);
}

console.log(factorial(5)); // 120

```

11. Prime Number Check

```
function isPrime(num) {
  if (num <= 1) return false;

  for (let i = 2; i < num; i++) {
    if (num % i === 0) return false;
  }

  return true;
}

console.log(isPrime(7)); // true
```

12. Debounce Function

```
function debounce(fn, delay) {
  let timer;

  return function (...args) {
    clearTimeout(timer);

    timer = setTimeout(() => {
      fn.apply(this, args);
    }, delay);
  };
}
```

13. Flatten Nested Array

```
function flattenArray(arr) {
  return arr.flat(Infinity);
}

console.log(flattenArray([1, [2, [3, 4]]]));
```

14. Find Second Largest Number

```
function secondLargest(arr) {
  const unique = [...new Set(arr)].sort((a, b) => b - a);
  return unique[1];
}

console.log(secondLargest([10, 5, 8, 20]));
```